LESSON 8.7

# Equations and Inequalities Problem Solving

**Start here** · Words, worked example, and sentence starters

**Name** _____ **Date** _____ **Period** _____

**LEARNING TARGET** I can model and solve real-world problems using equations and inequalities.

**WHY THIS MATTERS** Detective Santos is closing a complex case. She has collected equations and inequalities from different parts of the investigation. One clue says: 'The total value of stolen items divided equally among 4 accomplices gave each person $85.' Another says: 'The lookout earned less than $50.' She must model both clues mathematically and check if the answers are reasonable. Can you help?

START HERE

## Words for this lesson

Read each word before you begin. Say it out loud.

| WORD | WHAT IT MEANS | EXAMPLE |
|---|---|---|
| **Model** *Spanish: Modelar* | To show a real-life situation with an equation or inequality. | 'Twice a number is 18' → 2n = 18 — the equation models the words |
| **Equation** *Spanish: Ecuación* | A math sentence with an equal sign showing both sides are the same. | x + 8 = 15 — exactly one answer: x = 7 |
| **Inequality** *Spanish: Desigualdad* | A math sentence that compares two sides with <, >, ≤, or ≥. | x + 5 > 12 — many answers: x = 8, 9, 10, ... |
| **Reasonableness** *Spanish: Razonabilidad* | Checking if your answer makes sense. | If the problem asks for a number of people and you get x = −3, that is NOT reasonable |
| **Variable** *Spanish: Variable* | A letter that stands for a number that is unknown or can change. | “5 less than a number is 12” becomes x − 5 = 12, where x is the variable. |
| **Constraint** *Spanish: Restricción* | A limit in a problem that tells what values are allowed. | “You can spend at most $20” is a constraint: cost ≤ 20. |

HOW IT WORKS

## Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

**Clue: 'A total divided equally among 4 people gave each $85.'**

1. This has an exact total, so I use an equation. Let t be the total.
2. Divided among 4 to get 85 means t ÷ 4 = 85, so t = 85 × 4 = 340.
3. Check reasonableness: $340 split 4 ways is $85 each. That makes sense!

NOW YOU TRY

## Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

**Clue: 'The lookout earned less than $50.' Equation or inequality?**

1. _____
2. _____
3. _____

**Answer:** _____

SAY IT AND WRITE IT

## Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- I use an equation when the problem says _____, but I use an inequality when the problem says _____ because _____.

**Word bank** Model | Equation | Inequality | Reasonableness | Variable | Constraint | equation | inequality | equal | at least | at most | more than

WATCH OUT!

## A common mistake

A common mistake in Equations and Inequalities Problem Solving is matching the wrong model to a key phrase. For 'the getaway car held no more than 6 bags,' students often write the equation $b = 6$ instead of the inequality $b \leq 6$, because they see one number and assume it must be exact. The same mistake happens in reverse: for 'the reward split equally among 5 people gave each $40,' students sometimes write the inequality $5p \geq 40$ instead of the equation $5p = 40$, because 'split equally' sounds like it could mean a range. Before you submit, ask: does this phrase give one exact value (equation, =) or a limit/range (inequality, <, >, ≤, ≥)?

LESSON 8.7

# Equations and Inequalities Problem Solving

**Level 0** · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

**Name** _____ **Date** _____ **Period** _____

**LEARNING TARGET** I can model and solve real-world problems using equations and inequalities.

**WHY THIS MATTERS** Detective Santos is closing a complex case. She has collected equations and inequalities from different parts of the investigation. One clue says: 'The total value of stolen items divided equally among 4 accomplices gave each person $85.' Another says: 'The lookout earned less than $50.' She must model both clues mathematically and check if the answers are reasonable. Can you help?

HOW IT WORKS

## Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

**Clue: 'A total divided equally among 4 people gave each $85.'**

1. This has an exact total, so I use an equation. Let t be the total.
2. Divided among 4 to get 85 means t ÷ 4 = 85, so t = 85 × 4 = 340.
3. Check reasonableness: $340 split 4 ways is $85 each. That makes sense!

## PART 1 Understand the idea

models and meaning

**1** **Circle the letter of the best answer. Show how you know.**

Which model is correct for: 'A number divided by 6 equals 7'?

(A) n / 6 = 7
(B) 6n = 7
(C) n − 6 = 7
(D) n + 6 = 7

**STEPS TO TRY**

1. Spot the operation phrase: 'divided by 6'. That phrase gives you the symbol between n and 6.
2. Keep the sentence order — the number n comes first, then the division by 6, then 'equals 7'.
3. Write n / 6 on the left and 7 on the right. Which choice shows division rather than multiplying or adding 6?

SHOW YOUR WORK

## PART 2 Apply it

real-world problems

**2** **Circle the letter of the best answer. Show how you know.**

A museum has some paintings. After adding 12, they have 45. Which equation models this?

(A) p + 12 = 45

(B) p − 12 = 45

(C) 12p = 45

(D) p / 12 = 45

**STEPS TO TRY**

1. The museum STARTS with an unknown number of paintings p, then 12 more are ADDED to reach 45.
2. 'Adding 12' means the model joins p and 12 with a plus sign, and the total 45 goes after the equals sign.
3. Write the story in order: p, plus 12, equals 45. Which choice matches without switching the operation?

**SHOW YOUR WORK**

**3** **Circle the letter of the best answer. Show how you know.**

A detective needs more than 8 hours to finish the investigation. She has already worked 3 hours. Which inequality represents the additional hours h she needs?

(A) 3 + h > 8

(B) 3 + h = 8

(C) h > 3

(D) 8 + h > 3

**STEPS TO TRY**

1. There are two time pieces: the 3 hours already worked and the extra hours h. Together they must pass 8 hours.
2. 'More than 8' means the TOTAL of 3 + h sits on the big side of a strict inequality — not an equals sign.
3. Combine first: 3 + h. Then compare that total to 8 using the strict 'more than' symbol. Which choice is built that way?

**SHOW YOUR WORK**

**4** **Circle the letter of the best answer. Show how you know.**

A bus can carry at most 48 passengers. There are already 31 on board. Which inequality shows how many more can board?

(A) $31 + p \leq 48$

(B) $31 + p > 48$

(C) $31 + p = 48$

(D) $p - 31 \leq 48$

**STEPS TO TRY**

1. The bus already has 31 riders, and p more may board — but the TOTAL cannot pass 48.
2. 'At most 48' allows exactly 48 but nothing above it, so you need the symbol with the 'or equal' bar, not > or =.
3. Add the parts first: 31 + p. Then cap that sum at 48 with the 'at most' symbol. Which choice matches?

**SHOW YOUR WORK**

**EXPLAIN YOUR THINKING**

**How do you decide whether a word problem needs an equation or an inequality?**

**SENTENCE STARTER** I use an equation when the problem says ___, but I use an inequality when the problem says ___ because ___.

**3-Point Exit Checklist:** ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

**How did it go?** ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help